



Faculty of Engineering
Department of Biomedical Engineering

**System Requirements for Non-Contact Type Clinical
Thermometer**

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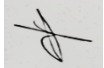
CHANGE RECORDS

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1. PURPOSE

The purpose of this report is to define exactly what the thermometer must do, how well it must do it, and under what constraints, ensuring that the final device is safe, accurate, reliable, and meets the intended user needs.

2. SCOPE

The first step in building a reliable clinical non-contact thermometer is figuring out exactly what it needs to do and how well it needs to do it. This means deciding on the infrared technology you'll use, which features are must-haves versus nice-to-haves and understanding the real-world conditions where the device will be used. You'll need to think through things like how accurate the readings should be, how far away the sensor can work, how quickly it responds, whether it needs calibration, what diagnostics to include, how users will interact with it, power requirements, and the physical design itself.

Getting these requirements nailed down early saves a lot of headaches later. It helps ensure the thermometer will work the way clinicians expect while also satisfying regulators. Industry standards are helpful here because they give you a framework for capturing what matters, along with proven approaches for designing the system. They also guide you in setting up the testing and validation work that proves your thermometer is safe and performs as intended.

3. REFERENCES

3.1. IEC 60601-1:2005+A1:2012: Medical electrical equipment - Part 1: General requirements for basic safety and essential performance

3.2. IEC 60601-1-2:2014: Medical electrical equipment — Part 1-2: General requirements for basic safety and essential performance — Collateral Standard: Electromagnetic disturbances Requirements and tests

3.3. ISO 80601-2-56:2017: Part 2-56: Particular requirements for basic safety and essential performance of clinical thermometers for body temperature measurement

3.4. ISO 9001: Quality management systems - Requirements

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4. REQUIREMENTS

4.1 PHYSICAL REQUIREMENTS

4.1.1 WEIGHT

The non-contact clinical thermometer will be 50 grams to 400 grams maximum.

4.1.2 SURFACE

The non-contact thermometer's surface will be smooth, durable and designed to be cleaned when needed. It must feel good on hand and people must feel the quality.

4.2. FUNCTIONAL REQUIREMENTS

4.2.1. OPERATING CONTROLS

The non-contact thermometer shall feature a minimal, user-friendly control interface (including elders), with no more than two buttons: a power button, a measurement button. Buttons shall be very easy to click and could be silent switch.

4.2.2. POWER SUPPLY

Supply should be provided from battery. The charging section should have battery production and must support 1A current. Charging section shall stop charging when battery has level of 100% charge and shall notify the user when its level reached full.

4.2.3. UNITS OF MEASURE

The non-contact thermometer will display the temperature in degrees Celsius °C

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4.2.4. RATED OUTPUT RANGE

The output temperature of the non-contact thermometer shall measure from 25,0 °C to 45,0 °C.

4.2.5. ACCURACY

The accuracy of the non-contact thermometer will provide an accuracy of ± 0.3 °C.

4.2.6. MEASUREMENT TIME

The measurement time shall be between 0 to 1 seconds.

4.2.7. ALARM CONDITIONS

Alarm conditions are optional. If the value is above acceptable range (more than 39.5 degrees Celsius) it must alert using a buzzer on board. This will express the situation's emergency.

4.3. ENVIRONMENTAL CONDITIONS

4.3.1. OPERATING TEMPERATURE

The thermometer should operate in standard conditions with a range of 15°C to 80°C

4.3.2. STORAGE TEMPERATURE

The thermometer should be suitable for storage at temperatures between -20°C and 90°C without damage.

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4.4. USABILITY

4.4.1. DISPLAY SIZE

Character size will be greater than 4 mm for users to read easily. Screen must be an OLED. If it is OLED display it shall be more than one colour and it shall be more than 0.96 inches. It shall work on <3.3 V DC voltage difference range. I2C communication protocol could be used or other favored communication protocol.

4.4.2. MEASURING SITE

The non-contact thermometer will have used and pointed on the forehead as a measuring site to read body temperature. To read the temperature correctly it should be within the distance of 1 to 5 cm from the forehead.

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